



06. Express + MongoDB Architecture Setup

Standard Project File Structure



Server Boilerplate Configuration

```
index.js - Server Setup

const express = require("express");
const app = express();
const path = require("path");
const mongoose = require("mongoose");
const methodOverride = require("method-override");
const Chat = require("../models/chat.js");

app.set("views", path.join(__dirname, "views"));
app.set("view engine", "ejs");
app.use(express.urlencoded({ extended: true }));
app.use(methodOverride("_method"));
```

Database Seeding (init.js)

A separate script is used to populate dummy records during development:

```
init.js - Seeding Script

Chat.insertMany([
  { from: "Neha", to: "Priya", msg: "Send notes!", created_at: new Date() },
  { from: "Rohit", to: "Mohit", msg: "Exam tomorrow", created_at: new Date() }
]);
```



Full RESTful CRUD Routes Implementation



RESTful Routes Architecture Matrix

Method & Path	Action	Mongoose Operation
GET /chats	Index (List all)	await Chat.find()
GET /chats/new	New (Show form)	res.render("new.ejs")
POST /chats	Create (Insert)	new Chat({ ... }).save()
GET /chats/:id/edit	Edit (Form)	await Chat.findById(id)
PUT /chats/:id	Update	await Chat.findByIdAndUpdate(id, ..., { runValidators: true })
DELETE /chats/:id	Destroy	await Chat.findByIdAndDelete(id)



Complete Route Implementations

```

index.js - REST Routes

// 1. INDEX: Show all chats
app.get("/chats", async (req, res) => {
  let chats = await Chat.find();
  res.render("index.ejs", { chats });
});

// 2. UPDATE: PUT /chats/:id
app.put("/chats/:id", async (req, res) => {
  let { id } = req.params;
  let { msg: newMsg } = req.body;
  await Chat.findByIdAndUpdate(id, { msg: newMsg }, { runValidators: true });
  res.redirect("/chats");
});

// 3. DESTROY: DELETE /chats/:id
app.delete("/chats/:id", async (req, res) => {
  let { id } = req.params;
  await Chat.findByIdAndDelete(id);
  res.redirect("/chats");
});

```